Riemann surfaces Problem set 3

1. For an irreducible nonconstant polynomial $P \in \mathbb{C}[z, w]$ let $C_P \subset \mathbb{C}^2, \overline{C_P} \subset \mathbb{CP}^2$ denote the corresponding affine and projective algebraic curves respectively, and let X_P denote the normalization of $\overline{C_P}$. Find the points at infinity and singular points of $\overline{C_P}$, and compute the genus of X_P in the following cases:

a) $P(z, w) = w^2 - Q(z)$, where $Q \in \mathbb{C}[z]$ is a polynomial of degree $2n$ for some $n \geq 1$, with all zeroes distinct.

b) $P(z, w) = w^2 - Q(z)$, where $Q \in \mathbb{C}[z]$ is a polynomial of degree $2n - 1$ for some $n \geq 1$, with all zeroes distinct.

c) $P(z, w) = w^2 - z^2(z - 1)$.

d) $P(z, w) = w^n + z^n - 1$, where $n \geq 2$.

e) $P(z, w) = w^3 - z(z - 1)$.

2. Show that there is a lattice $L \subset \mathbb{C}$ and there are nonconstant, L-periodic meromorphic functions f, g on $\mathbb{C}$ such that $f^3 + g^3 = 1$.

3. Let X, Y be compact Riemann surfaces of genus $g(X), g(Y)$ respectively, and suppose there is a nonconstant holomorphic map $f : X \to Y$. Show that $g(X) \geq g(Y)$.

4. Let K be a finite extension of $\mathbb{C}(z)$. Show that there is a compact Riemann surface X such that $\mathcal{M}(X)$ is isomorphic to K as $\mathbb{C}$-algebras.

5. Let $f : X \to \mathbb{C}$ be a nonconstant meromorphic function on a connected Riemann surface X. Show that the meromorphic 1-form df satisfies $\mathrm{ord}_p(df) = \deg_p(f) - 1$ if p is not a pole of f, and $\mathrm{ord}_p(df) = \mathrm{ord}_p(f) - 1$ if p is a pole of f.

6. Let f be a nonconstant meromorphic function on a compact Riemann surface X of genus $g \geq 0$. Show that

$$\sum_{p \in X} \mathrm{ord}_p(df) = 2g - 2.$$

7. Let ω_1, ω_2 be linearly independent meromorphic 1-forms on a Riemann surface X.

a) Show that there is a nonconstant meromorphic function f on X such that $\omega_2 = f \cdot \omega_1$.

b) Show that if X is a compact Riemann surface of genus two then the meromorphic function f has degree two as a branched cover of $\hat{\mathbb{C}}$.

8. Let $P(z, w) = w^2 - Q(z)$ where Q is a polynomial of degree 3 with distinct roots.

a) Show that the projective algebraic curve $\overline{C_P}$ is nonsingular.

b) Let $x, y \in \mathcal{M}(\overline{C_P})$ be the meromorphic functions induced by the coordinate functions on $\mathbb{C}^2$. Show that the meromorphic 1-form $\omega = dx/y$ is in fact a holomorphic 1-form which is nowhere zero.